MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

LATCH, DIGITAL, KEY CODE GDJ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1 1 Scope This document establishes the requirements and test procedures for the acquisition of an electronic latch module

* 1 2 Classification This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer)

2 2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3 2 Correlation requirements Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.153 failures per million operating hours for class I or 0.439 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GDJ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A

3.5.6 Depth. The module shall have no additional depth requirements

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/1-1 for class I and M28787/1-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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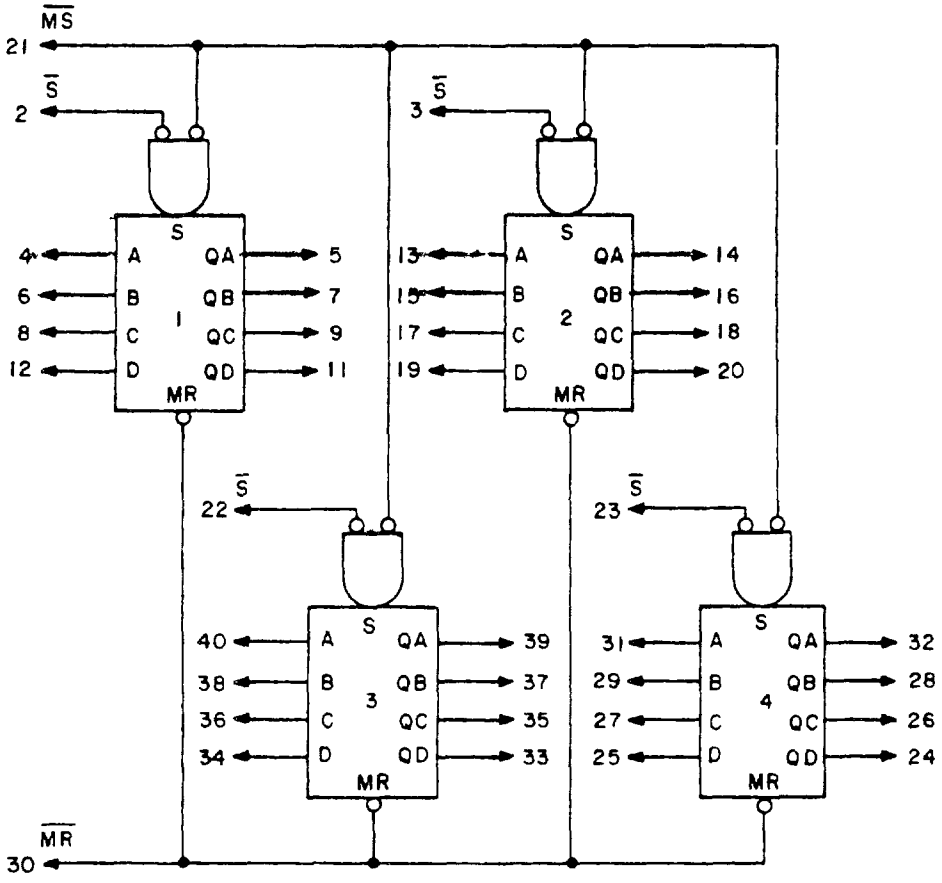


FIGURE 1 GDJ module function diagram, latch

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* TABLE I Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A16	V_{OUT} (low-loaded)	III	S	0 01	0 80	0 01	0 80	0.01	0.80	0.01	0 80	0.01	0 80	V
B1-B22	Input breakdown current	IV	S	0 0	1 0	0 0	1 0	0 0	1.0	0 0	1.0	0.0	1 0	mA
C1-C22	Input clamp diode voltage	V	S	-0.4	-1 5	-0 4	-1 5	-0 4	-1 5	-0 4	-1 5	-0 4	-1.5	V
D1-D16	Input current, high level (data)	VI	X	0	90	0	90	0	90	0	90	0	90	µA
D17-D20	Input current, high level (strobe)	VI	X	0	60	0	60	0	60	0	60	0	60	µA
D21,D22	Input current, high level (master strobe and reset)	VI	X	0	240	0	240	0	240	0	240	0	240	µA
E1-E16	Input current, low level (data)	VI	X	0 10	2.73	0.10	2 73	0 10	2.73	0 10	2 73	0.10	2 73	mA
E17-E20	Input current, low level (strobe)	VI	X	0 05	1 73	0.05	1 73	0 05	1.73	0 05	1 73	0 05	1.73	mA
E21,E22	Input current, low level (master strobe and reset)	VI	X	0 10	6 92	0.10	6 92	0 10	6 92	0 10	6 92	0.10	6.92	mA
F1-F16	Output short circuit current	VI	X	10	70	10	70	10	70	10	70	10	70	mA
G1-G48	Output voltage, high level	VII	X	2 4	4 5	2 4	4 5	2 4	4.5	2.4	4.5	2 4	4 5	V
H1-H64	Output voltage, low level													
	Initial	VII	X	0 00	0 40	0 00	0.40	0 00	0 40	0 00	0.40	0 00	0 40	V
	End-of-life	VII	X	0 00	0 40	0 00	0 40	0.00	0 40	0 00	0.44	0 00	0 44	V
J	Power supply current	VIII	X	50	230	50	230	50	230	50	230	50	230	mA
K1-K16	Transition time (t_{TLH})													
	Initial	IX	S	5	55	5	60	5	60	5	60	5	60	ns
	End-of-life	IX	S	5	60	5	60	5	60	5	60	5	60	ns

See footnotes at end of table

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L1-L16	Transition time (t_{THL})													
	Initial	IX	S	5	25	5	30	5	30	5	30	5	30	ns
	End-of-life	IX	S	5	30	5	30	5	30	5	30	5	30	ns
M1-M16	Propagation delay time (t_{PLH}) strobe to output													
	Initial	IX	S	5	45	5	50	5	50	5	50	5	50	ns
	End-of-life	IX	S	5	50	5	50	5	50	5	50	5	50	ns
M17-M32	Propagation delay time (t_{PLH}), data to output													
	Initial	IX	S	5	35	5	40	5	40	5	40	5	40	ns
	End-of-life	IX	S	5	40	5	40	5	40	5	40	5	40	ns
N1-N16	Propagation delay time (t_{PHL}), strobe to output													
	Initial	IX	S	5	35	5	40	5	40	5	40	5	40	ns
	End-of-life	IX	S	5	40	5	40	5	40	5	40	5	40	ns
N17-N32	Propagation delay time (t_{PHL}), data to output													
	Initial	IX	S	5	35	5	40	5	40	5	40	5	40	ns
	End-of-life	IX	S	5	40	5	40	5	40	5	40	5	40	ns
P1-P16	Setup time check, data to strobe	X	S	35	65	35	65	35	65	35	65	35	65	ns
Q1-Q16	Hold time check, data to strobe	X	S	35	65	35	65	35	65	35	65	35	65	ns
P1-P16	Master reset delay time													
	Initial	XI	S	5	35	5	40	5	40	5	40	5	40	ns
	End-of-life	XI	S	5	40	5	40	5	40	5	40	5	40	ns
R17-R32	Master reset delay time (PWC)													
	Initial	XI	S	400	600	400	600	400	600	400	600	400	600	ns
	End-of-life	XI	S	400	600	400	600	400	600	400	600	400	600	ns

See footnotes at end of table

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* TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S1-S16	Master reset recovery time check	XII	S	180	220	180	220	180	220	180	220	180	220	ns
T	Power supply transient rise time	XIII	S	2.5	7.5	2.5	7 5	2 5	7 5	2 5	7.5	2 5	7 5	µs
U	Isolation													
	Initial	XIV	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	XIV	S	1 0	∞	1.0	∞	1 0	∞	1.0	∞	1 0	∞	MΩ
V1-V16	Output crosstalk, high level	XV	Q	0.0	0.4	0.0	0.4	0.0	0 4	0 0	0.4	0 0	0 4	V
W1-W16	Output crosstalk, low level	XV	Q	0.0	1.080	0.0	1 080	0.0	1 080	0 0	1.080	0 0	1 080	V
W17-W64	Output crosstalk, low level	XV	Q	0.0	0.480	0.0	0.480	0.0	0.480	0.0	0.480	0.0	0 480	V
X1-X16	Input capacitance, data	XVI	Q	2	15	2	15	2	15	2	15	2	15	pF
X17-X20	Input capacitance, strobe	XVI	Q	3	22	3	22	3	22	3	22	3	22	pF
X21,X22	Input capacitance, master strobe and reset	XVI	Q	8	60	8	60	8	60	8	60	8	60	pF
Y	Power supply transient amplitude	XVII	Q	0	600	0	600	0	600	0	600	0	600	mV

1/ "X" parameters shall be 100 percent tested during module acceptance tests

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only

4/ Unless otherwise specified, limits are for initial and end-of-life

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4 4.1 Alternate test methods Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If test conditions A, B, F, or U are part of a sequence of tests, the order of testing shall be as follows U, A, B, F where applicable, followed by the remaining tests in the sequence.

4 5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4 5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through T, V, W, and Y shall be performed in the order specified in 4 4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5 1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787 During the vibration tests, the module shall be operated in the test circuit shown on figure 4 Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition T, shall be performed using the initial limits at 25° ±5°C ambient.

4.5 3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through T, V, W, and Y shall be performed in the order specified in 4 4 2 using the operating temperature end-of-life requirements of table I Following the return of the module to 25°C, electrical tests C through T and V through Y shall be performed in the order specified in 4 4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787

6 NOTES

6 1 Definitions. Definitions are as specified in MIL-STD-1331 and as follows.

6.1.1 Data setup time. Data setup time is the time a data signal must be present on the input before the strobe transition from low to high occurs, in order for the latch to recognize and store the new data.

6.1.2 Data hold time. Data hold time is the time a data signal must be present on the input after the strobe transition from low to high occurs, in order for the latch to recognize and store the new data.

6.1 3 Master reset recovery time. Master reset recovery time is the time that the strobe must remain low after the master reset transition from low to high occurs, in order for the latch to recognize and store high data.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6 3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4)

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* 6.4 Cross reference of classification. Military specification MIL-M-28787/1 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658643. Part number M28787/1-1 supersedes part numbers 2658643 and M28787/1. These parts are interchangeable.

* 6.5 Module description The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

	Class I	Class II	Unit
Supply voltage	5.0 ±10%	5.0 ±10%	Vdc
Maximum power dissipation (outputs not loaded)	1.265	1.265	W
Input voltage			
Low level	0.8 max	0.8 max	V
High level	2.0 min	2.0 min	V
Input current			
Low level (V_{IL} = 0 V)			
A, B, C, D inputs	-2.73 max	-2.73 max	mA
Strobe inputs	-1.73 max	-1.73 max	mA
Master strobe input	-6.92 max	-6.92 max	mA
Master reset input	-6.92 max	-6.92 max	mA
Input current			
High level (V_{IH} = 2.4 V)			
A, B, C, D inputs	90 max	90 max	µA
Strobe inputs	60 max	60 max	µA
Master strobe input	240 max	240 max	µA
Master reset input	240 max	240 max	µA
Output voltage			
Low level	0.40 max	0.44 max	V
High level	2.4 min	2.4 min	V
Output drive current maximum capability			
Low level (V_{OL} = 0.4 V)	14.4	14.4	mA
High level (V_{OH} = 2.4 V)	-0.6	-0.6	mA
Speed (strobe)	DC to 20 MHz	DC to 20 MHz	
Pulse width			
Strobe	20 min	20 min	ns
Master reset	20 min	20 min	ns
Propagation delay time (t_{PLH})			
Strobe to output	5 min 50 max	5 min 50 max	ns
Data to output	5 min 40 max	5 min 40 max	ns
Propagation delay time (t_{PHL})			
Strobe to output	5 min 40 max	5 min 40 max	ns
Data to output	5 min 40 max	5 min 40 max	ns
Setup time (data to strobe)			
Setup (L)	18 min	18 min	ns
Setup (H)	18 min	18 min	ns
Hold time (data to strobe)			
Hold (L)	15 max	15 max	ns
Hold (H)	15 max	15 max	ns
Master reset delay time	40 max	40 max	ns
Master reset recovery time	12 min	12 min	ns

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* 6.6 Module function. This module contains four 4-input storage latches with four corresponding true outputs (Q). Each 4-bit latch has one strobe input ANDed with a master strobe input. All four latches are controlled by one master strobe input and one master reset input.

* 6.6.1 Data entry. Data can be entered into the latch when both the individual strobe ($\overline{S}$) and the master strobe ($\overline{MS}$) inputs are low. As long as this logic conditin exists, the output of the latch will follow the input. If either of the strobe inputs ($\overline{MS}$ or $\overline{S}$) goes high, the data present in the latch at the time is held in the latch and is no longer affected by the data input. The master reset ($\overline{MR}$) overrides all other input conditions and forces the outputs of all the latches low when a low signal is applied to the master reset input.

* 6.6.2 Input to output relationship. Each latch has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/-5/

$\overline{MR}$	$\overline{MS}$	$\overline{S}$	Data	Qn	Operation
H	L	L	L	L	Data entry
H	L	L	H	H	Data entry
H	L	H	X	Qn-1	Hold
H	H	L	X	Qn-1	Hold
H	H	H	X	Qn-1	Hold
L	X	X	X	L	Reset

1/ X = Don't care.
2/ L = Low-voltage level
3/ H = High-voltage level
4/ Qn-1 = Previous output state.
5/ Qn = Present output state

* 6.6.3 Parallel outputs. Paralleling of outputs is not permissible

* 6.6.4 Crosstalk relationships. The module has the followng crosstalk relationships.

* 6.6.4.1 Module crosstalk. When connected in the worst case mode of operation (all outputs but one switching simultaneously from a high level to a low level), the GDJ module could exhibit substantial crosstalk. When the quiet line is a low level, there may be as much as an 1,080 mV positive going spike coincident with the negative edge of the switching outputs.

* 6 6.4.2 Circuit crosstalk. The crosstalk on any output pin is nearly directly proportional to the number of switching outputs. For every switching output in circuit 1, the noise induced on a low level quiet output in circuit 1 is as much as 120 mV. For every switching output outside circuit 1 the noise induced on a low level quiet output in circuit 1 is as much as 60 mV. The preceding relationships also hold for circuits 2-4.

6 7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

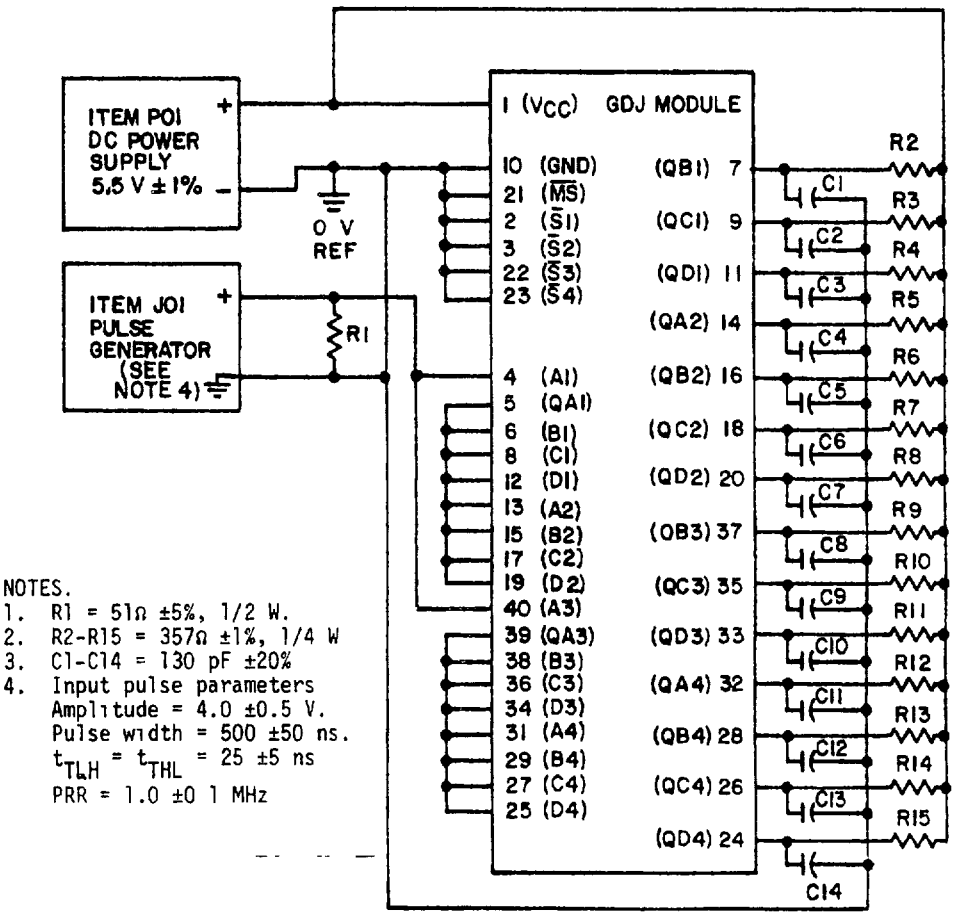
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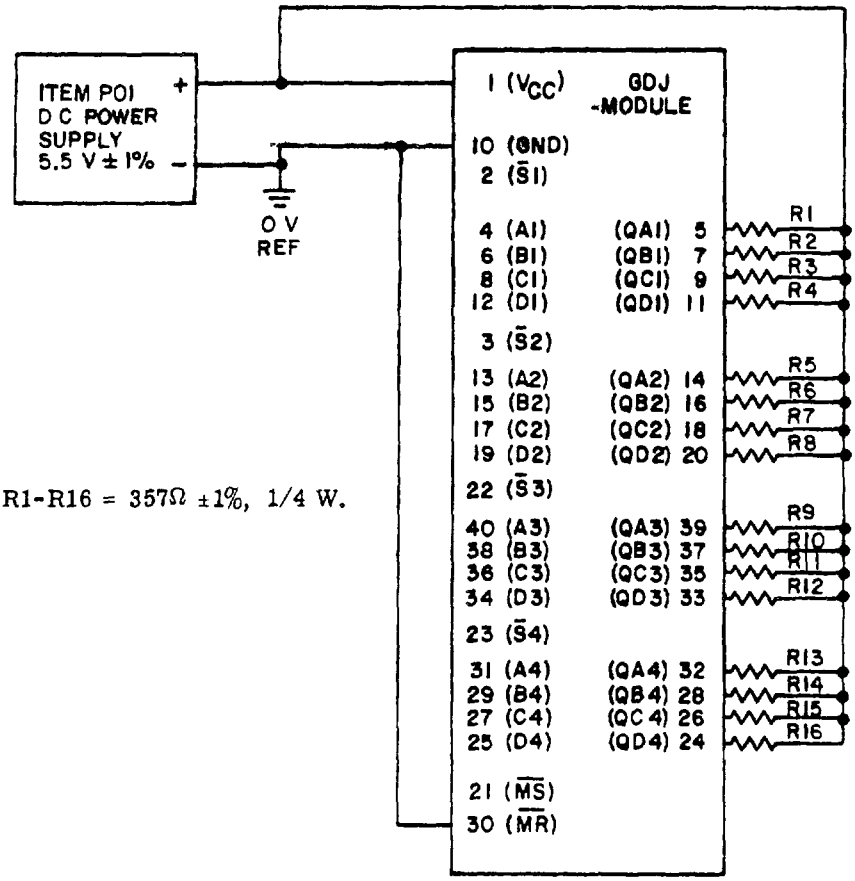
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NOTES.
1. R1 = 51Ω ±5%, 1/2 W.
2. R2-R15 = 357Ω ±1%, 1/4 W
3. C1-C14 = 130 pF ±20%
4. Input pulse parameters
   Amplitude = 4.0 ±0.5 V.
   Pulse width = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns
   PRR = 1.0 ±0.1 MHz

FIGURE 2. Operating temperature and life test circuit.

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NOTE R1-R16 = 357Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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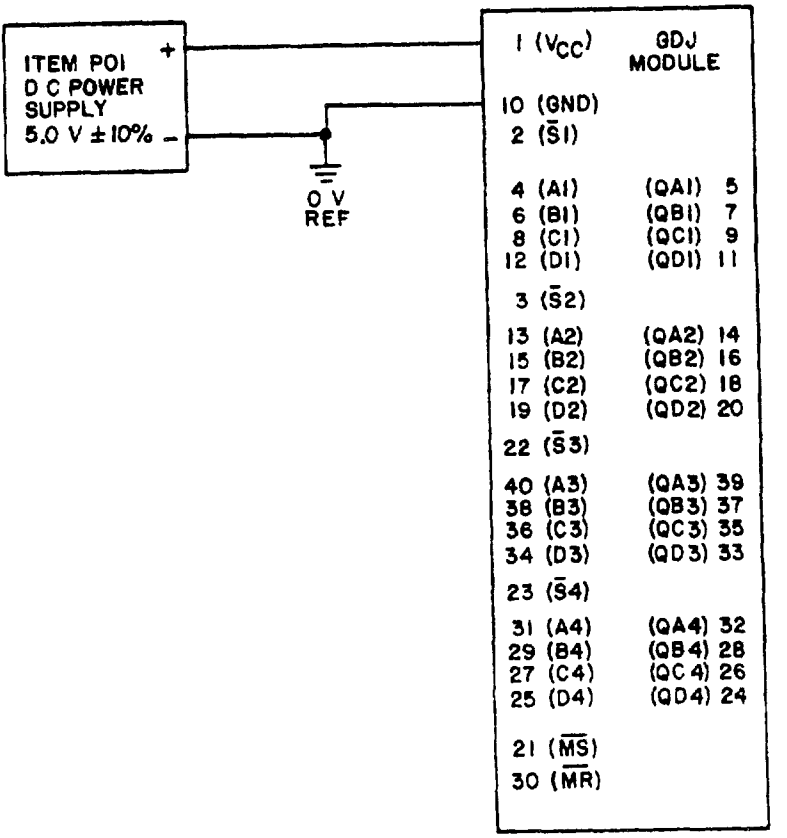


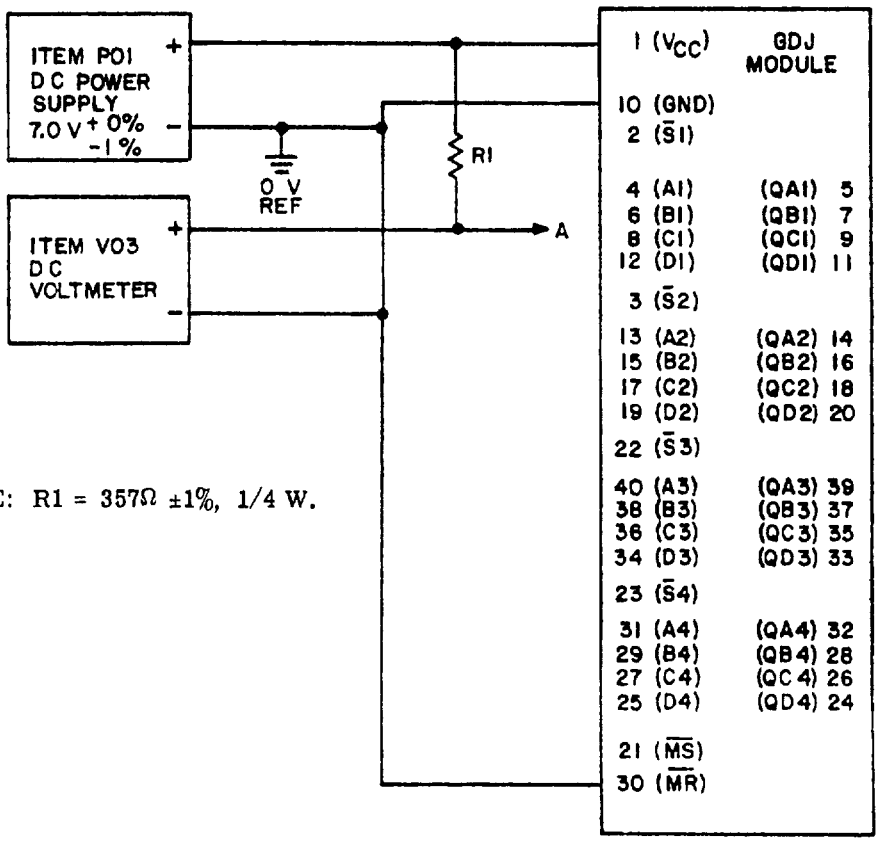
FIGURE 4. Vibration test circuit

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TABLE III V_{OUT} (low-loaded) test

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	QA1	V_{OUT} (low-loaded)	5	0.01	0.80	V	5
A2	QB1		7				
A3	QC1		9				
A4	QD1		11				
A5	QA2		14				
A6	QB2		16				
A7	QC2		18				
A8	QD2		20				
A9	QA3		39				
A10	QB3		37				
A11	QC3		35				
A12	QD3		33				
A13	QA4		32				
A14	QB4		28				
A15	QC4		26				
A16	QD4		24				

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NOTE: R1 = 357Ω ±1%, 1/4 W.

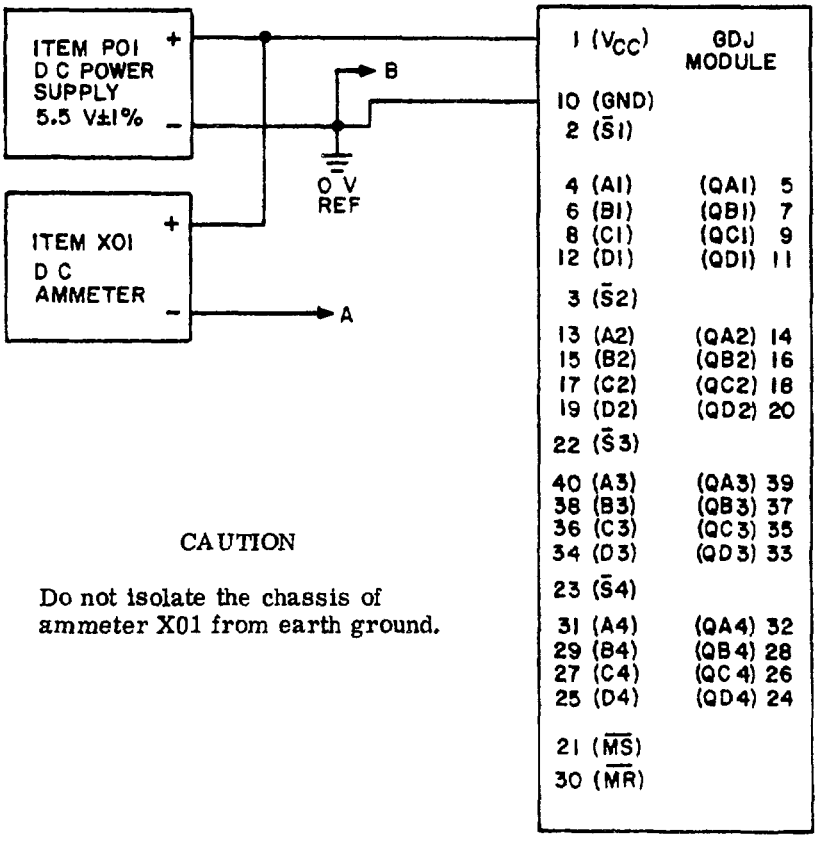
FIGURE 5. V_{OUT} (low-loaded) test circuit for table III

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TABLE IV Input breakdown current

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	A1	Input breakdown current	4	30	0.0	1.0	mA	6
B2	B1		6					
B3	C1		8					
B4	D1		12					
B5	A2		13					
B6	B2		15					
B7	C2		17					
B8	D2		19					
B9	A3		40					
B10	B3		38					
B11	C3		36					
B12	D3		34					
B13	A4		31					
B14	B4		29					
B15	C4		27					
B16	D4		25					
B17	$\overline{S}1$		2	21				
B18	$\overline{S}2$		3					
B19	$\overline{S}3$		22					
B20	$\overline{S}4$		23					
B21	$\overline{MS}$		21	2, 3, 22, 23				
B22	$\overline{MR}$		30	Open				

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

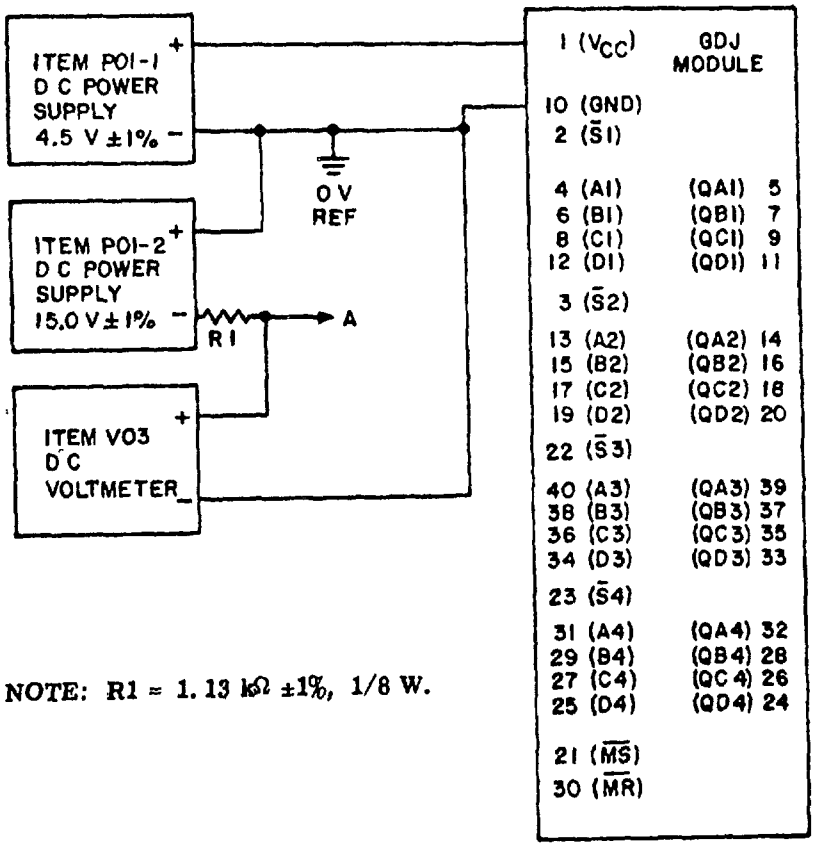
FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V Input clamp diode voltage test

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	A1 Input clamp diode voltage	4	-0.4	-1.5	V	7
C2	B1	6				
C3	C1	8				
C4	D1	12				
C5	A2	13				
C6	B2	15				
C7	C2	17				
C8	D2	19				
C9	A3	40				
C10	B3	38				
C11	C3	36				
C12	D3	34				
C13	A4	31				
C14	B4	29				
C15	C4	27				
C16	D4	25				
C17	$\overline{S}1$	2				
C18	$\overline{S}2$	3				
C19	$\overline{S}3$	22				
C20	$\overline{S}4$	23				
C21	$\overline{MS}$	21				
C22	$\overline{MR}$	30				

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NOTE: R1 = 1.13 kΩ ±1%, 1/8 W.

FIGURE 7. Input clamp diode voltage test circuit for table V.

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TABLE VI. Input current (high-level and low-level) and output short circuit current test.

Test condition		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	A1	Input current, high level	4	Open	30	0	90	µA	8
D2	B1		6						
D3	C1		8						
D4	D1		12						
D5	A2		13						
D6	B2		15						
D7	C2		17						
D8	D2		19						
D9	A3		40						
D10	B3		38						
D11	C3		36						
D12	D3		34						
D13	A4		31						
D14	B4		29						
D15	C4		27						
D16	D4		25						
D17	$\overline{S}1$		2		21		60		
D18	$\overline{S}2$		3						
D19	$\overline{S}3$		22						
D20	$\overline{S}4$		23						
D21	$\overline{MS}$		21		2,3,22,23		240		
D22	$\overline{MR}$		30		Open		240		

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TABLE VI. Input current (high-level and low-level) and output short circuit current test - Continued.

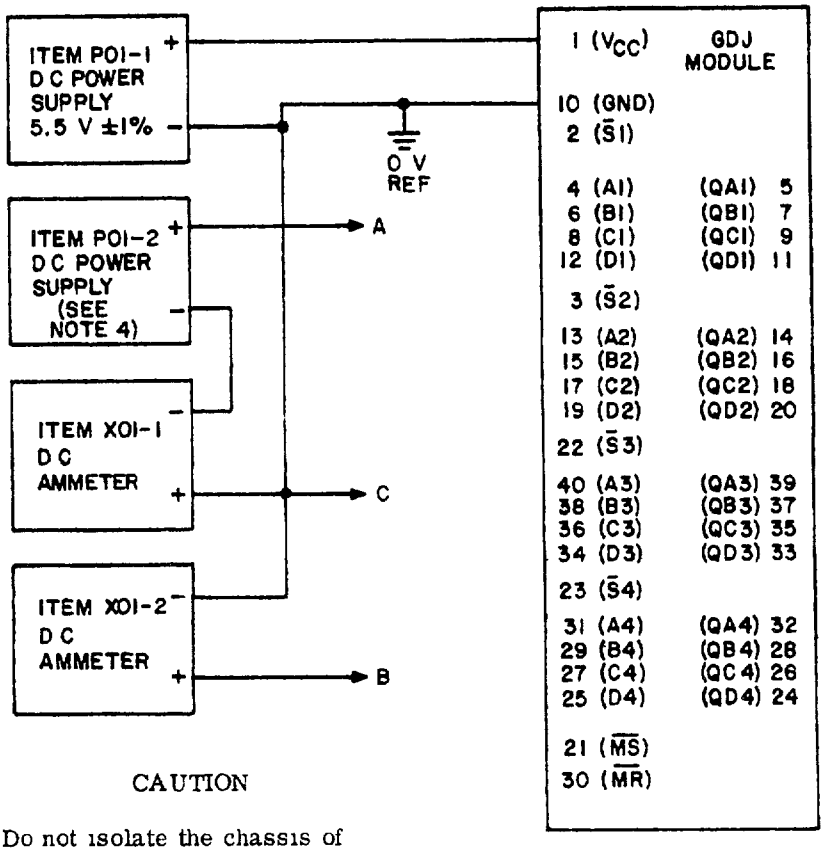
Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	A1	Input current, low level	Open	4	2, 21	0.10	2.73	mA	8
E2	B1			6					
E3	C1			8					
E4	D1			12					
E5	A2			13	3, 21				
E6	B2			15					
E7	C2			17					
E8	D2			19					
E9	A3			40	21, 22				
E10	B3			38					
E11	C3			36					
E12	D3			34					
E13	A4			31	21, 23				
E14	B4			29					
E15	C4			27					
E16	D4			25					
E17	$\overline{S}1$		21	2	Open	0.05	1.73		
E18	$\overline{S}2$			3					
E19	$\overline{S}3$			22					
E20	$\overline{S}4$			23					
E21	$\overline{MS}$		2,3,22,23	21		0.10	6.92		
E22	$\overline{MR}$		Open	30		0.10	6.92		

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TABLE VI. Input current (high-level and low-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	QA1	Output short circuit current	4	5	2, 21	10	70	mA	8
F2	QB1		6	7					
F3	QC1		8	9					
F4	QD1		12	11					
F5	QA2		13	14	3, 21				
F6	QB2		15	16					
F7	QC2		17	18					
F8	QD2		19	20					
F9	QA3		40	39	21, 22				
F10	QB3		38	37					
F11	QC3		36	35					
F12	QD3		34	33					
F13	QA4		31	32	21, 23				
F14	QB4		29	28					
F15	QC4		27	26					
F16	QD4		25	24					

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES.
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. With point A disconnected, ammeter X01-1 should indicate zero current on the 1.0 μA range.
3. When performing output short circuit current test, only one output shall be shorted at a time and the measurement taken within 5 seconds.
4. Adjust item P01-2 to 2.4 V ±1% for test condition D and to 4.5 V ±1% for test conditions E and F.

FIGURE 8. Input current (high-level and low-level) and output short circuit current test circuit for table VI.

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TABLE VII. Output voltage (high-level and low-level) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G1	QA1 Output voltage, high level	4	5	Open	Open	2, 21	2.4	4.5	V	9
G2	QB1	6	7							
G3	QC1	8	9							
G4	QD1	12	11							
G5	QA2	13	14			3, 21				
G6	QB2	15	16							
G7	QC2	17	18							
G8	QD2	19	20							
G9	QA3	40	39			21, 22				
G10	QB3	38	37							
G11	QC3	36	35							
G12	QD3	34	33							
G13	QA4	31	32			21, 23				
G14	QB4	29	28							
G15	QC4	27	26							
G16	QD4	25	24							
G17	QA1	2, 21	5	30	4	Open				
G18	QB1		7		6					
G19	QC1		9		8					
G20	QD1		11		12					

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TABLE VII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G21	QA2 Output voltage, high level	3, 21	14	30	13	Open	2.4	4.5	V	9
G22	QB2		16		15					
G23	QC2		18		17					
G24	QD2		20		19					
G25	QA3	21, 22	39		40					
G26	QB3		37		38					
G27	QC3		35		36					
G28	QD3		33		34					
G29	QA4	21, 23	32		31					
G30	QB4		28		29					
G31	QC4		26		27					
G32	QD4		24		25					
G33	QA1	30	5	Open	4	2, 21				
G34	QB1		7		6					
G35	QC1		9		8					
G36	QD1		11		12					
G37	QA2		14		13	3, 21				
G38	QB2		16		15					
G39	QC2		18		17					
G40	QD2		20		19					

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TABLE VII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G41	QA3	Output voltage, high level	30	39	Open	40	21,22	2.4	4.5	V	9
G42	QB3			37		38					
G43	QC3			35		36					
G44	QD3			33		34					
G45	QA4			32		31	21, 23				
G46	QB4			28		29					
G47	QC4			26		27					
G48	QD4			24		25					
H1	QA1	Output voltage, low level	4	5		Open	2, 21	0.00	0 40		
H2	QB1		6	7							
H3	QC1		8	9							
H4	QD1		12	11							
H5	QA2		13	14			3, 21				
H6	QB2		15	16							
H7	QC2		17	18							
H8	QD2		19	20							
H9	QA3		40	39			21, 22				
H10	QB3		38	37							
H11	QC3		36	35							
H12	QD3		34	33							

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TABLE VII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect. A to pin	Connect. B to pin	Connect. C to pin	Connect. D to pin	Connect. E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H13	QA4 Output voltage, low level	31	32	Open	Open	21, 23	0.00	0.40	V	9
H14	QB4	29	28							
H15	QC4	27	26							
H16	QD4	25	24							
H17	QA1	2	5	30	4	21				
H18	QB1		7		6					
H19	QC1		9		8					
H20	QD1		11		12					
H21	QA2	3	14		13					
H22	QB2		16		15					
H23	QC2		18		17					
H24	QD2		20		19					
H25	QA3	22	39		40					
H26	QB3		37		38					
H27	QC3		35		36					
H28	QD3		33		34					
H29	QA4	23	32		31					
H30	QB4		28		29					
H31	QC4		26		27					
H32	QD4		24		25					

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TABLE VII. Output voltage (high-level and low-level) test - Continued.

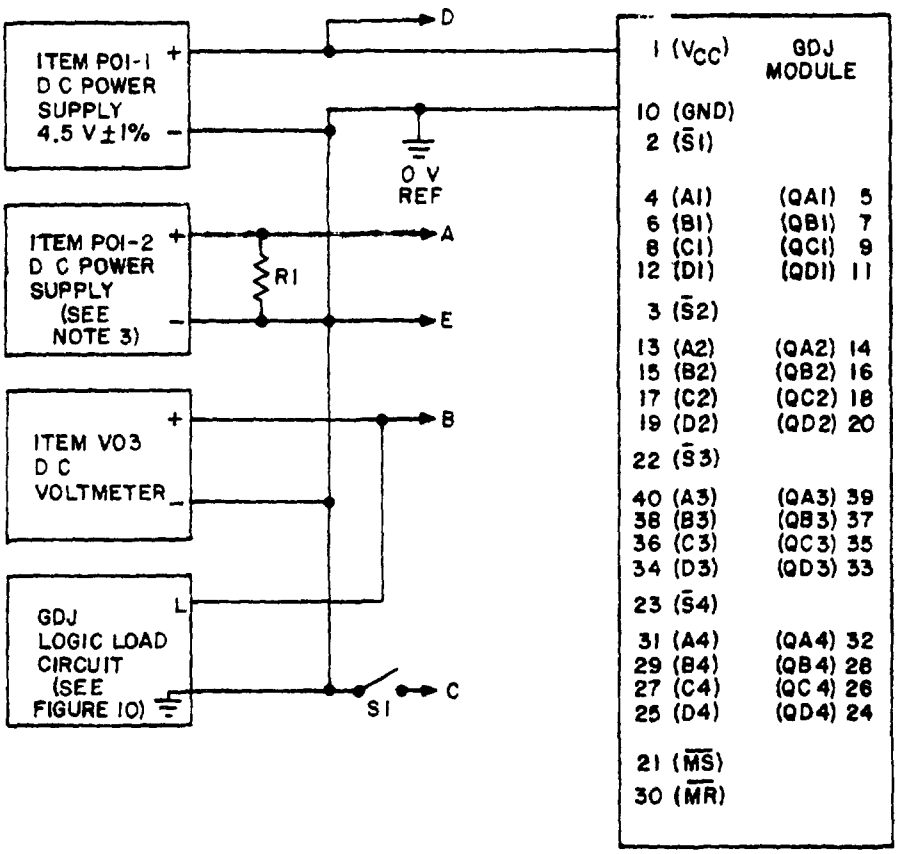
Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
H33	QA1	Output voltage, low level	21	5	30	4	2	0.00	0.40	V	9
H34	QB1			7		6					
H35	QC1			9		8					
H36	QD1			11		12					
H37	QA2			14		13	3				
H38	QB2			16		15					
H39	QC2			18		17					
H40	QD2			20		19					
H41	QA3			39		40	22				
H42	QB3			37		38					
H43	QC3			35		36					
H44	QD3			33		34					
H45	QA4			32		31	23				
H46	QB4			28		29					
H47	QC4			26		27					
H48	QD4			24		25					
H49	QA1		30	5	Open	4	2, 21				
H50	QB1			7		6					
H51	QC1			9		8					
H52	QD1			11		12					

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TABLE VII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
H53	QA2	Output voltage, low level	30	14	Open	13	3, 21	0.00	0.40	V	9
H54	QB2			16		15					
H55	QC2			18		17					
H56	QD2			20		19					
H57	QA3			39		40	21,22				
H58	QB3			37		38					
H59	QC3			35		36					
H60	QD3			33		34					
H61	QA4			32		31	21, 23				
H62	QB4			28		29					
H63	QC4			26		27					
H64	QD4			24		25					

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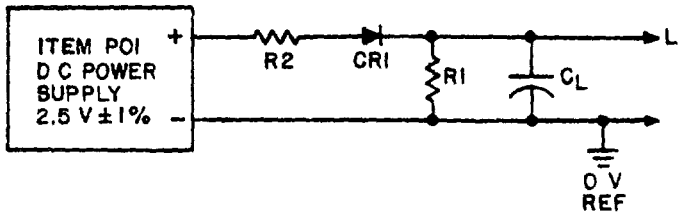
NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. S1 = SPST switch.
3. Adjust item P01-2 as follows.

Test condition	Voltage
G1-G16, G33-G48, H17-H48	2.0 V ±1%
G17-G32, H1-H16, H49-H64	0.8 V ±1%

4. After making all circuit connections, close switch S1 for a minimum of 100 ns, then open switch S1 and allow a minimum of 50 ns before recording measurement.

FIGURE 9. Output voltage (high-level and low-level) test circuit for table VII.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. C_L = 130 ±5 pF (including probe and parasitic capacitance) for test conditions K, L, M, N, P, R, and S.
3. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions V and W.
4. R1 = 4.02 kΩ ±1%, 1/8 W.
5. R2 = Select value for a current flow of 15.6 mA ±1% from point L to ground.
6. Capacitor C_L is not required for dc electrical test conditions G and H.
7. During output switching, item P01 shall have a maximum overshoot of 500 mV_{pk}.

FIGURE 10. GDJ logic load circuit.

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TABLE VIII. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
J	Power supply current	50	230	mA	11

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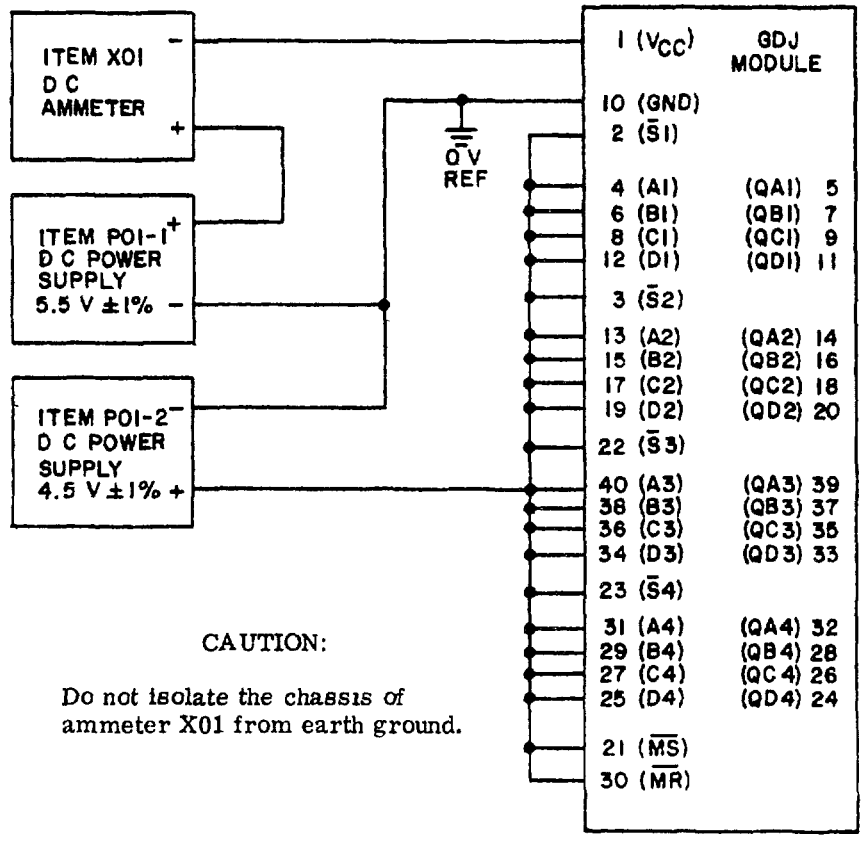


FIGURE 11. Power supply current test circuit for table VIII.

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
K1	QA1 Transition time (t_{TLH})	4	21	5	2	Open	5	55	ns	12, 13
K2	QB1	6		7						
K3	QC1	8		9						
K4	QD1	12		11						
K5	QA2	13		14	3					
K6	QB2	15		16						
K7	QC2	17		18						
K8	QD2	19		20						
K9	QA3	40		39	22					
K10	QB3	38		37						
K11	QC3	36		35						
K12	QD3	34		33						
K13	QA4	31		32	23					
K14	QB4	29		28						
K15	QC4	27		26						
K16	QD4	25		24						
L1	QA1 Transition time (t_{THL})	4		5	2			25		
L2	QB1	6		7						
L3	QC1	8		9						
L4	QD1	12		11						

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TABLE IX Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
L5	QA2 Transition time (t_{THL})	13	21	14	3	Open	5	25	ns	12, 13
L6	QB2	15		16						
L7	QC2	17		18						
L8	QD2	19		20						
L9	QA3	40		39	22					
L10	QB3	38		37						
L11	QC3	36		35						
L12	QD3	34		33						
L13	QA4	31		32	23					
L14	QB4	29		28						
L15	QC4	27		26						
L16	QD4	25		24						
M1	$\overline{MS}$ to QA1 Propagation delay time (t_{PLH})	4		5	2			45		
M2	$\overline{MS}$ to QB1	6		7						
M3	$\overline{MS}$ to QC1	8		9						
M4	$\overline{MS}$ to QD1	12		11						
M5	$\overline{MS}$ to QA2	13		14	3					
M6	$\overline{MS}$ to QB2	15		16						
M7	$\overline{MS}$ to QC2	17		18						
M8	$\overline{MS}$ to QD2	19		20						

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TABLE IX Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M9	$\overline{MS}$ to QA3 Propagation delay time (t_{PLH})	40	21	39	22	Open	5	45	ns	12, 13
M10	$\overline{MS}$ to QB3	38		37						
M11	$\overline{MS}$ to QC3	36		35						
M12	$\overline{MS}$ to QD3	34		33						
M13	$\overline{MS}$ to QA4	31		32	23					
M14	$\overline{MS}$ to QB4	29		28						
M15	$\overline{MS}$ to QC4	27		26						
M16	$\overline{MS}$ to QD4	25		24						
M17	A1 to QA1	4	Open	5	2, 21			35		
M18	B1 to QB1	6		7						
M19	C1 to QC1	8		9						
M20	D1 to QD1	12		11						
M21	A2 to QA2	13		14	3, 21					
M22	B2 to QB2	15		16						
M23	C2 to QC2	17		18						
M24	D2 to QD2	19		20						
M25	A3 to QA3	40		39	21, 22					
M26	B3 to QB3	38		37						
M27	C3 to QC3	36		35						
M28	D3 to QD3	34		33						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M29	A4 to QA4 Propagation delay time (t_{PLH})	31	Open	32	21, 23	Open	5	35	ns	12, 13
M30	B4 to QB4	29		28						
M31	C4 to QC4	27		26						
M32	D4 to QD4	25		24						
N1	$\overline{MS}$ to QA1 Propagation delay time (t_{PHL})	4	21	5	2			45		
N2	$\overline{MS}$ to QB1	6		7						
N3	$\overline{MS}$ to QC1	8		9						
N4	$\overline{MS}$ to QD1	12		11						
N5	$\overline{MS}$ to QA2	13		14	3					
N6	$\overline{MS}$ to QB2	15		16						
N7	$\overline{MS}$ to QC2	17		18						
N8	$\overline{MS}$ to QD2	19		20						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N9	$\overline{MS}$ to QA3 Propagation delay time (t_{PHL})	40	21	39	22	Open	5	35	ns	12, 13
N10	$\overline{MS}$ to QB3	38		37						
N11	$\overline{MS}$ to QC3	36		35						
N12	$\overline{MS}$ to QD3	34		33						
N13	$\overline{MS}$ to QA4	31		32	23					
N14	$\overline{MS}$ to QB4	29		28						
N15	$\overline{MS}$ to QC4	27		26						
N16	$\overline{MS}$ to QD4	25		24						
N17	A1 to QA1	4	Open	5	2, 21					
N18	B1 to QB1	6		7						
N19	C1 to QC1	8		9						
N20	D1 to QD1	12		11						
N21	A2 to QA2	13		14	3, 21					
N22	B2 to QB2	15		16						
N23	C2 to QC2	17		18						
N24	D2 to QD2	19		20						
N25	A3 to QA3	40		39	21, 22					
N26	B3 to QB3	38		37						
N27	C3 to QC3	36		35						
N28	D3 to QD3	34		33						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N29	A4 to QA4 Propagation delay time (t_{PHL})	31	Open	32	21, 23	Open	5	35	ns	12, 13
N30	B4 to QB4	29		28						
N31	C4 to QC4	27		26						
N32	D4 to QD4	25		24						

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TABLE X. Setup time check and hold time check test.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P1	A1 to $\overline{S1}$	Setup time check	4	2	5	21	Open	35	65	ns	12, 14
P2	B1 to $\overline{S1}$		6		7						
P3	C1 to $\overline{S1}$		8		9						
P4	D1 to $\overline{S1}$		12		11						
P5	A2 to $\overline{S2}$		13	3	14						
P6	B2 to $\overline{S2}$		15		16						
P7	C2 to $\overline{S2}$		17		18						
P8	D2 to $\overline{S2}$		19		20						
P9	A3 to $\overline{S3}$		40	22	39						
P10	B3 to $\overline{S3}$		38		37						
P11	C3 to $\overline{S3}$		36		35						
P12	D3 to $\overline{S3}$		34		33						
P13	A4 to $\overline{S4}$		31	23	32						
P14	B4 to $\overline{S4}$		29		28						
P15	C4 to $\overline{S4}$		27		26						
P16	D4 to $\overline{S4}$		25		24						
Q1	A1 to $\overline{S1}$	Hold time check	4	2	5						
Q2	B1 to $\overline{S1}$		6		7						
Q3	C1 to $\overline{S1}$		8		9						
Q4	D1 to $\overline{S1}$		12		11						

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TABLE X. Setup time check and hold time check test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q5	A2 to $\overline{S2}$ Hold time check	13	3	14	21	Open	35	65	ns	12, 14
Q6	B2 to $\overline{S2}$	15		16						
Q7	C2 to $\overline{S2}$	17		18						
Q8	D2 to $\overline{S2}$	19		20						
Q9	A3 to $\overline{S3}$	40	22	39						
Q10	B3 to $\overline{S3}$	38		37						
Q11	C3 to $\overline{S3}$	36		35						
Q12	D3 to $\overline{S3}$	34		33						
Q13	A4 to $\overline{S4}$	31	23	32						
Q14	B4 to $\overline{S4}$	29		28						
Q15	C4 to $\overline{S4}$	27		26						
Q16	D4 to $\overline{S4}$	25		24						

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TABLE XI Master reset delay time test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R1	Master reset delay time	2	30	5	21	4,6,8,12	5	35	ns	12, 15
R2				7						
R3				9						
R4				11						
R5		3		14		13,15, 17,19				
R6				16						
R7				18						
R8				20						
R9		22		39		40,38, 36,34				
R10				37						
R11				35						
R12				33						
R13		23		32		31,29, 27,25				
R14				28						
R15				26						
R16				24						

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TABLE XI. Master reset delay time test - Continued.

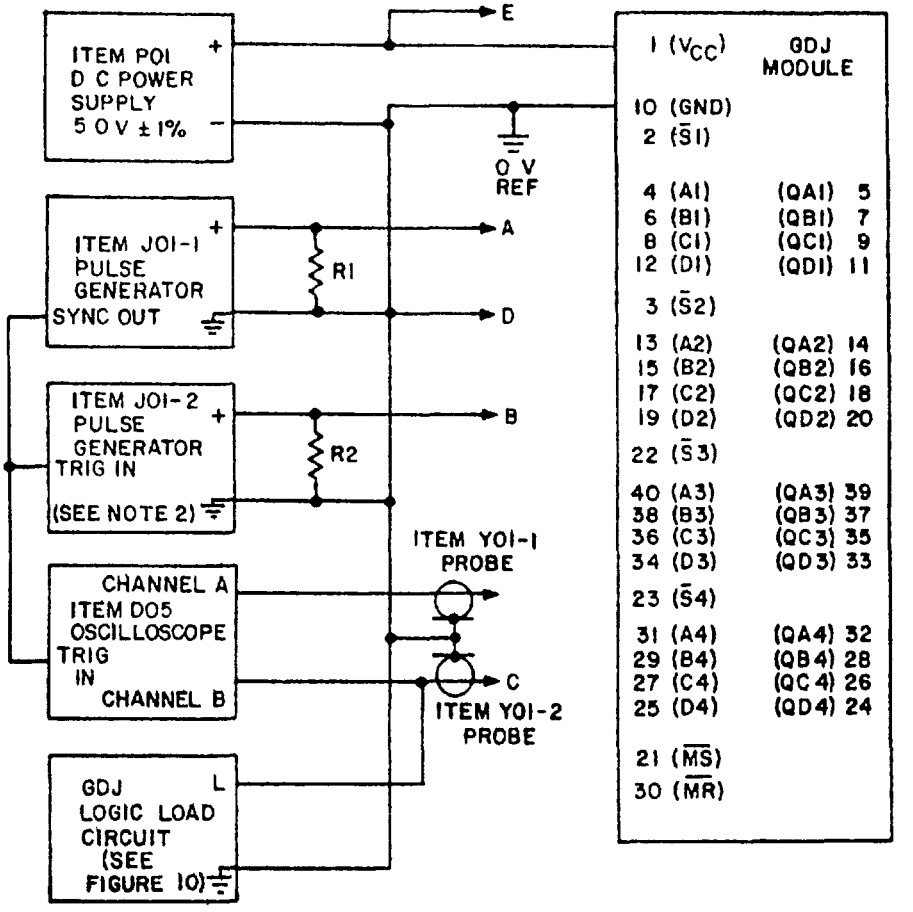
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R17	Master reset delay time (PWC)	2	30	5	21	4,6,8,12	400	600	ns	12, 15
R18				7						
R19				9						
R20				11						
R21		3		14		13,15, 17,19				
R22				16						
R23				18						
R24				20						
R25		22		39		40,38, 36,34				
R26				37						
R27				35						
R28				33						
R29		23		32		31,29, 27,25				
R30				28						
R31				26						
R32				24						

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TABLE XII Master reset recovery time check test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S1	Master reset recovery time check	21	30	5	2	Open	180	220	ns	12, 16
S2				7						
S3				9						
S4				11						
S5				14	3					
S6				16						
S7				18						
S8				20						
S9				39	22					
S10				37						
S11				35						
S12				33						
S13				32	23					
S14				28						
S15				26						
S16				24						

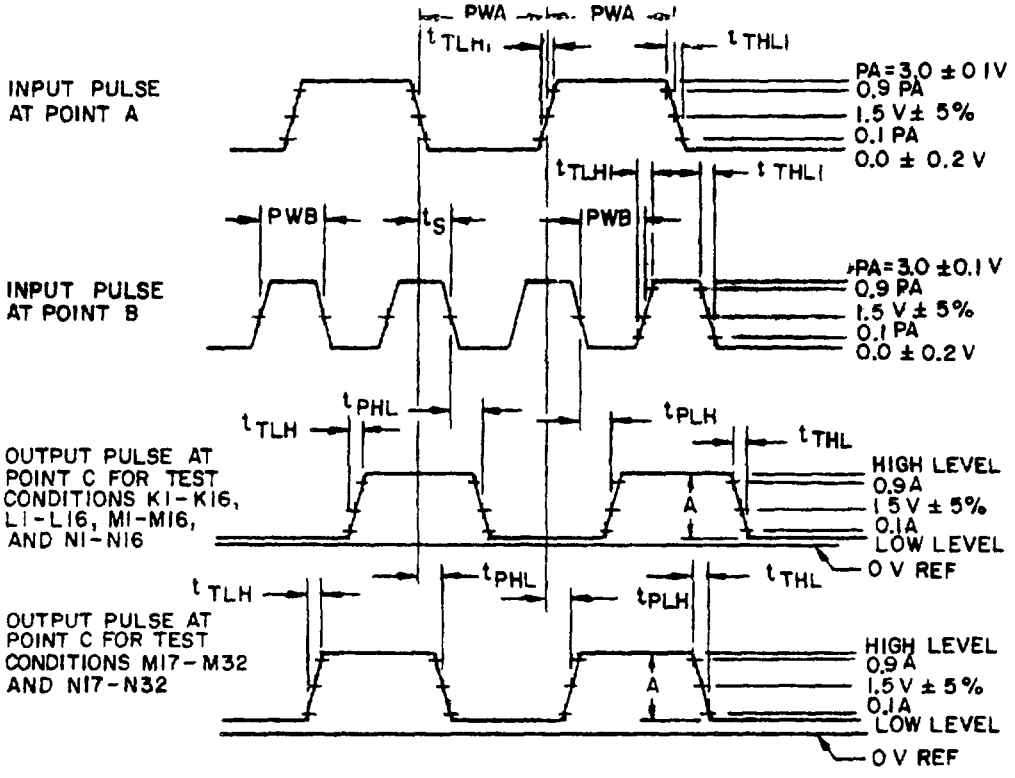
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:20Z
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NOTES
1. R1 = R2 = 51Ω ±5%, 1/2 W
2 Set item J01-2 to double pulse mode except for test conditions R1-R32

FIGURE 12 Transition time, (t_{TLH} and t_{THL}) propagation delay time, (t_{PLH} and t_{PHL}) setup time check, hold time check, master reset delay time, and master reset recovery time check test circuit for tables IX, X, XI, and XII

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NOTES

1 Input pulse parameters

PWA = 500 ±50 ns

PWB = 250 ±25 ns

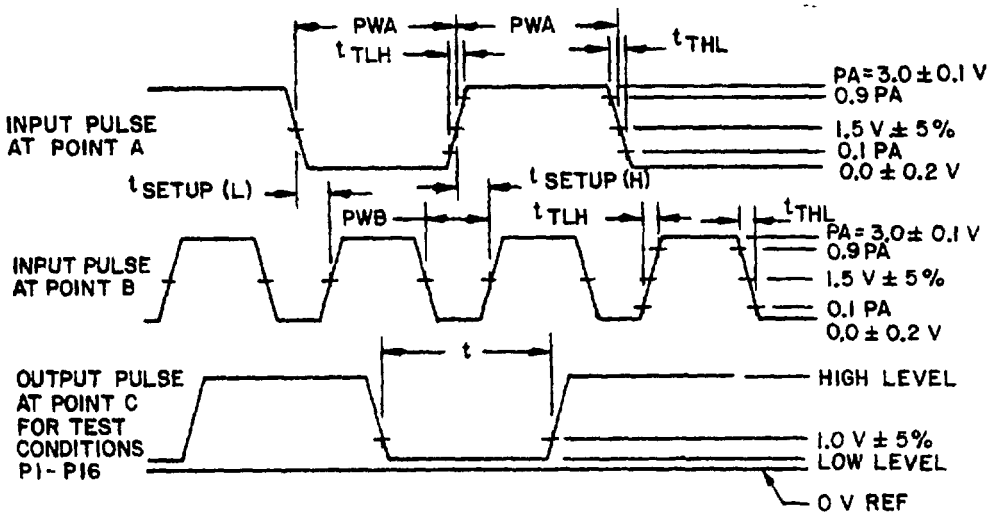
$t_{TLH1} = t_{THL1} = 10 \pm 2$ ns

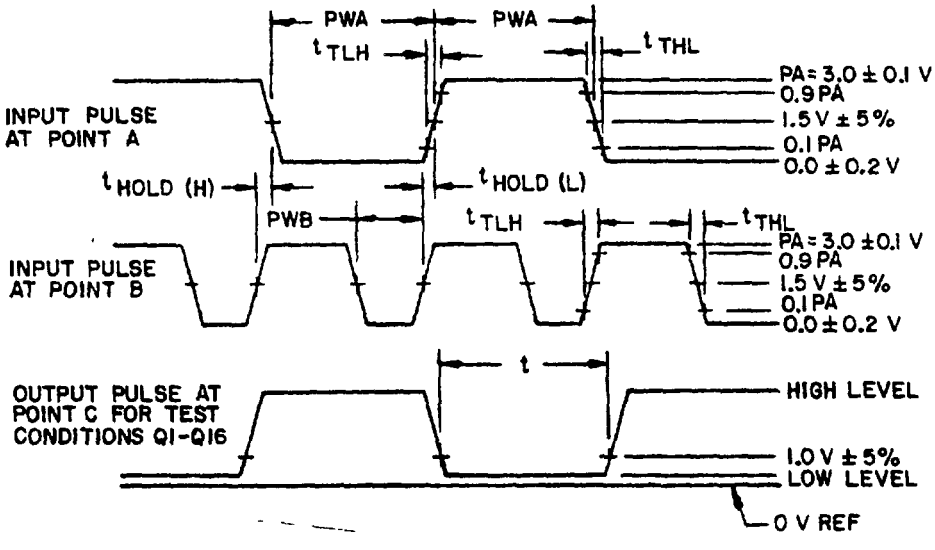
$t_S = 125 \pm 12$ ns

2 Before making the output pulse measurements, the output levels should be verified to be 0 4 V maximum for the low level and 2 4 minimum for the high level.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table IX

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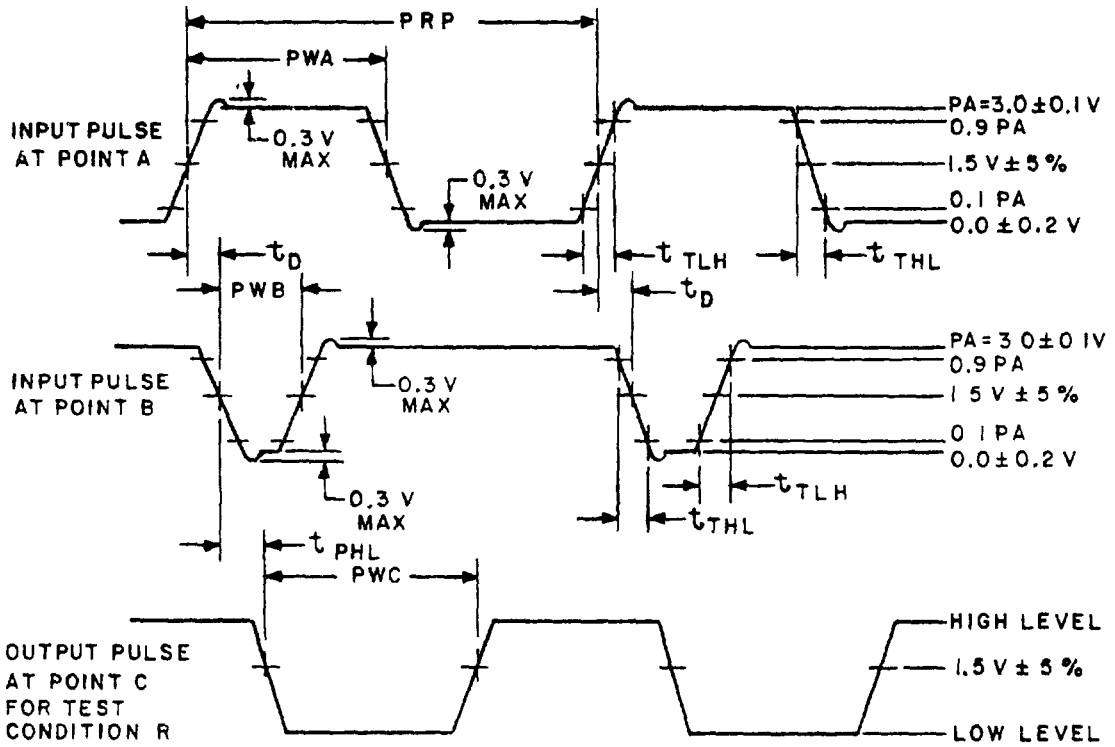




NOTE Input pulse parameters
PWA = 50 ±5 ns.
PWB = 20 ±2 ns.
$t_{TLH} = t_{THL} = 6 \pm 1$ ns
$t_{SETUP\ (L)} = t_{SETUP\ (H)} = 18 \pm 2$ ns
$t_{HOLD\ (L)} = t_{HOLD\ (H)} = 15 \pm 1$ ns.

* FIGURE 14. Setup time check and hold time check test circuit waveforms for table X.

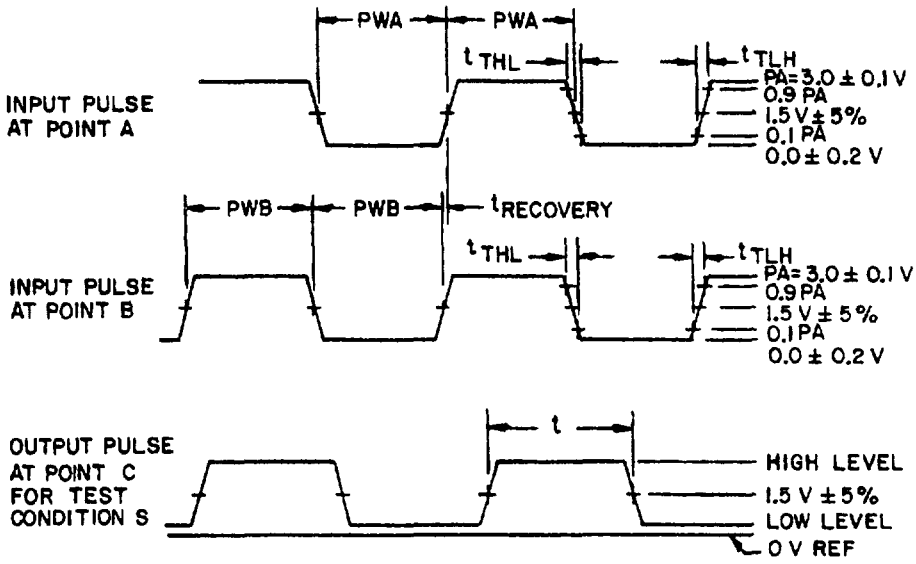
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NOTES
1. Input pulse parameters
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns
   t_D = 50 ±5 ns
   PWB = 30 ±3 ns
   $t_{TLH} = t_{THL}$ = 6 ±1 ns
2. Input at point B is taken from the negative output of item J01-2. Adjust the pulse level for a low level of 0.0 ±0.2 V.

FIGURE 15. Master reset delay time test circuit waveforms for table XI.

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NOTE: Input pulse parameters:
PWA = PWB = 200 ±20 ns.
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.
$t_{RECOVERY} = 16 \pm 2$ ns.

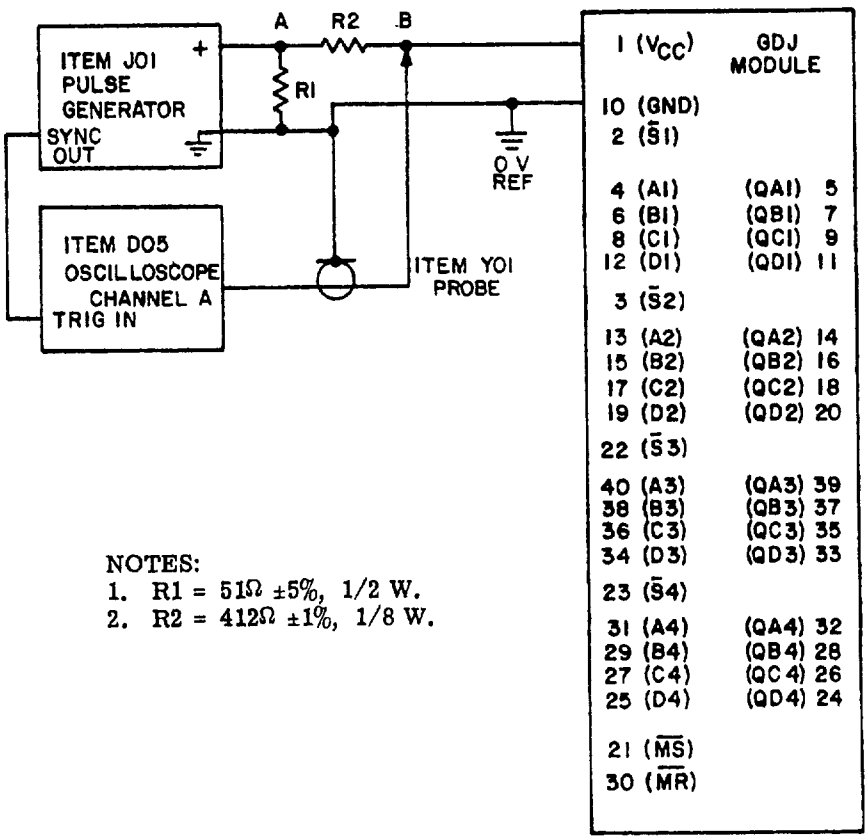
FIGURE 16. Master reset recovery time check test circuit waveforms for table XII.

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TABLE XIII. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient rise time	2.5	7.5	µs	17, 18

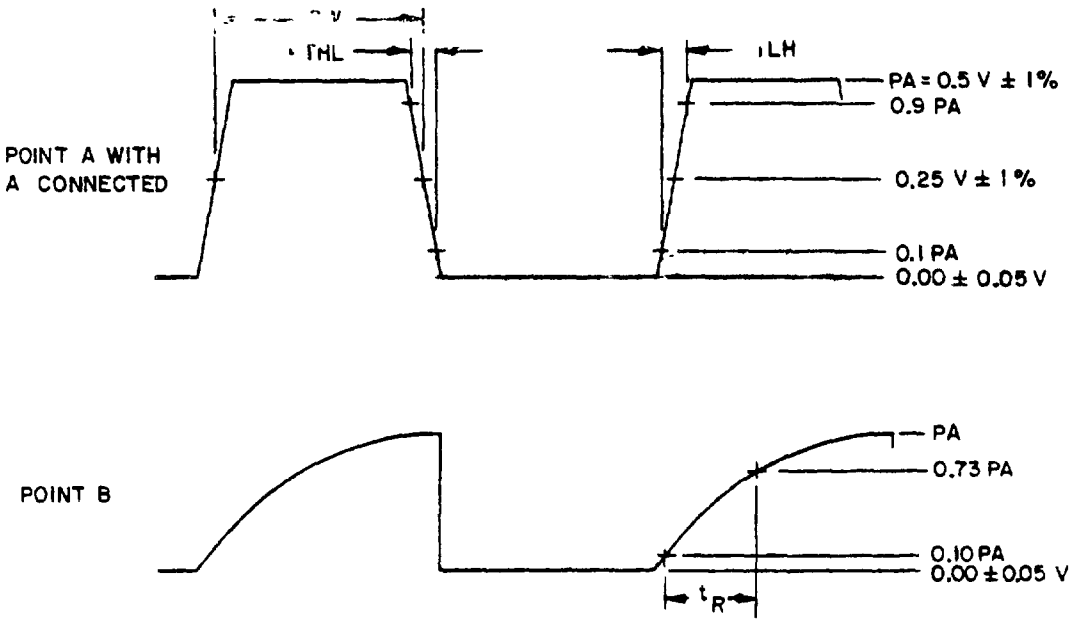
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:20Z
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 412Ω ±1%, 1/8 W.

FIGURE 17. Power supply transient rise time test circuit for table XIII.

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NOTE Input pulse parameters:
PW = 30 μs ±10%.
PRP = 60 μs ±10%.
$t_{TLH} = t_{THL}$ = 50 ns ±10%.

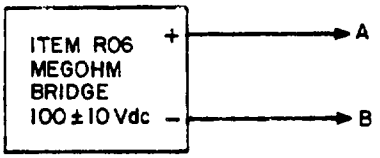
FIGURE 18 Power supply transient rise time test circuit waveforms for table XIII.

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TABLE XIV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U	Isolation	All pins	Frame	10	∞	MΩ	19

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WARNING.

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

	GDJ MODULE	
1 (V_{CC})		
10 (GND)		
2 ($\bar{S}1$)		
4 (A1)	(QA1)	5
6 (B1)	(QB1)	7
8 (C1)	(QC1)	9
12 (D1)	(QD1)	11
3 ($\bar{S}2$)		
13 (A2)	(QA2)	14
15 (B2)	(QB2)	16
17 (C2)	(QC2)	18
19 (D2)	(QD2)	20
22 ($\bar{S}3$)		
40 (A3)	(QA3)	39
38 (B3)	(QB3)	37
36 (C3)	(QC3)	35
34 (D3)	(QD3)	33
23 ($\bar{S}4$)		
31 (A4)	(QA4)	32
29 (B4)	(QB4)	28
27 (C4)	(QC4)	26
25 (D4)	(QD4)	24
21 ($\overline{MS}$)		
30 ($\overline{MR}$)		

FIGURE 19. Isolation test circuit for table XIV

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TABLE XV. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	QA1 Output crosstalk, high level	4,30	Open	6,8,12,13,15,17,19,25, 27,29,31,34,36,38,40	5	0.0	0.4	V	20,21
V2	QB1	6,30		4,8,12,13,15,17,19,25, 27,29,31,34,36,38,40	7				
V3	QC1	8,30		4,6,12,13,15,17,19,25, 27,29,31,34,36,38,40	9				
V4	QD1	12,30		4,6,8,13,15,17,19,25, 27,29,31,34,36,38,40	11				
V5	QA2	13,30		4,6,8,12,15,17,19,25, 27,29,31,34,36,38,40	14				
V6	QB2	15,30		4,6,8,12,13,17,19,25, 27,29,31,34,36,38,40	16				
V7	QC2	17,30		4,6,8,12,13,15,19,25, 27,29,31,34,36,38,40	18				
V8	QD2	19,30		4,6,8,12,13,15,17,25, 27,29,31,34,36,38,40	20				
V9	QA3	40,30		4,6,8,12,13,15,17,19,25, 27,29,31,34,36,38	39				
V10	QB3	38,30		4,6,8,12,13,15,17,19,25, 27,29,31,34,36,40	37				
V11	QC3	36,30		4,6,8,12,13,15,17,19,25, 27,29,31,34,38,40	35				
V12	QD3	34,30		4,6,8,12,13,15,17,19,25, 27,29,31,36,38,40	33				
V13	QA4	31,30		4,6,8,12,13,15,17,19,25, 27,29,34,36,38,40	32				
V14	QB4	29,30		4,6,8,12,13,15,17,19,25, 27,31,34,36,38,40	28				
V15	QC4	27,30		4,6,8,12,13,15,17,19, 25,29,31,34,36,38,40	26				
V16	QD4	25,30		4,6,8,12,13,15,17,19, 27,29,31,34,36,38,40	24				

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TABLE XV. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	x		
W1	QA1 Output crosstalk, low level	30	4	6,8,12,13,15,17,19,25, 27,29,31,34,36,38,40	5	0.0	1.080	V	20,21
W2	QB1		6	4,8,12,13,15,17,19,25, 27,29,31,34,36,38,40	7				
W3	QC1		8	4,6,12,13,15,17,19,25, 27,29,31,34,36,38,40	9				
W4	QD1		12	4,6,8,13,15,17,19,25, 27,29,31,34,36,38,40	11				
W5	QA2		13	4,6,8,12,15,17,19,25, 27,29,31,34,36,38,40	14				
W6	QB2		15	4,6,8,12,13,17,19,25, 27,29,31,34,36,38,40	16				
W7	QC2		17	4,6,8,12,13,15,19,25, 27,29,31,34,36,38,40	18				
W8	QD2		19	4,6,8,12,13,15,17,25, 27,29,31,34,36,38,40	20				
W9	QA3		40	4,6,8,12,13,15,17,19,25, 27,29,31,34,36,38	39				
W10	QB3		38	4,6,8,12,13,15,17,19,25, 27,29,31,34,36,40	37				
W11	QC3		36	4,6,8,12,13,15,17,19,25, 27,29,31,34,38,40	35				
W12	QD3		34	4,6,8,12,13,15,17,19,25, 27,29,31,36,38,40	33				
W13	QA4		31	4,6,8,12,13,15,17,19,25, 27,29,34,36,38,40	32				
W14	QB4		29	4,6,8,12,13,15,17,19,25, 27,31,34,36,38,40	28				
W15	QC4		27	4,6,8,12,13,15,17,19, 25,29,31,34,36,38,40	26				
W16	QD4		25	4,6,8,12,13,15,17,19, 27,29,31,34,36,38,40	24				

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TABLE XV. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W17	Output crosstalk, low level	30	4,6,8,12,13,15,17,19	25,27,29,31,34,36,38,40	5	0.0	0.480	V	20,21
W18			4,6,8,12,13,15,17,19	25,27,29,31,34,36,38,40	7				
W19			4,6,8,12,13,15,17,19	25,27,29,31,34,36,38,40	9				
W20			4,6,8,12,13,15,17,19	25,27,29,31,34,36,38,40	11				
W21					14				
W22					16				
W23					18				
W24					20				
W25			4,6,8,12,34,36,38,40	13,15,17,19,25,27,29,31	5				
W26					7				
W27					9				
W28					11				
W29					39				
W30					37				
W31					35				
W32					33				

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TABLE XV. Output crosstalk (high-level and low-level) test - Continued.

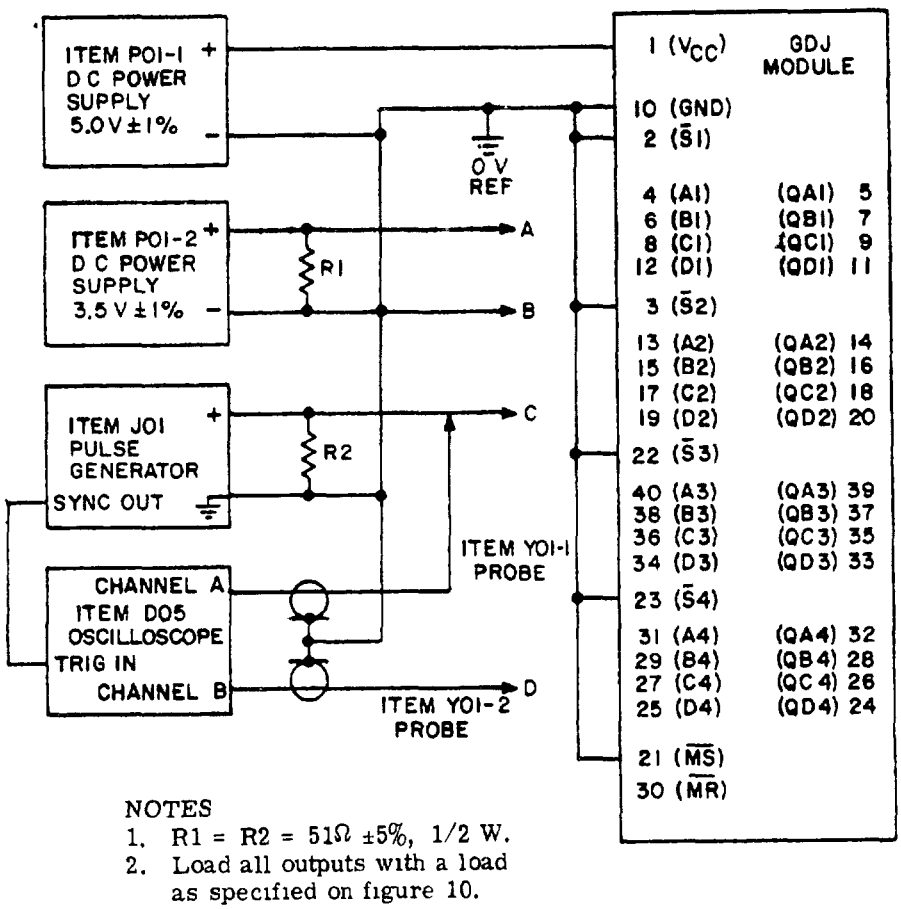
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W33	Output crosstalk, low level	30	4,6,8,12,25,27,29,31	13,15,17,19,34,36,38,40	5	0.0	0.480	V	20,21
W34					7				
W35					9				
W36					11				
W37					32				
W38					28				
W39					26				
W40					24				
W41			13,15,17,19,34,36,38,40	4,6,8,12,25,27,29,31	14				
W42					16				
W43					18				
W44					20				
W45					39				
W46					37				
W47					35				
W48					33				

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TABLE XV. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W49	Output crosstalk, low level	30	13,15,17,19,25,27,29,31	4,6,8,12,34,36,38,40	14	0.0	0.480	V	20,21
W50					16				
W51					18				
W52					20				
W53					32				
W54					28				
W55					26				
W56					24				
W57			25,27,29,31,34,36,38,40	4,6,8,12,13,15,17,19	39				
W58					37				
W59					35				
W60					33				
W61					32				
W62					28				
W63					26				
W64					24				

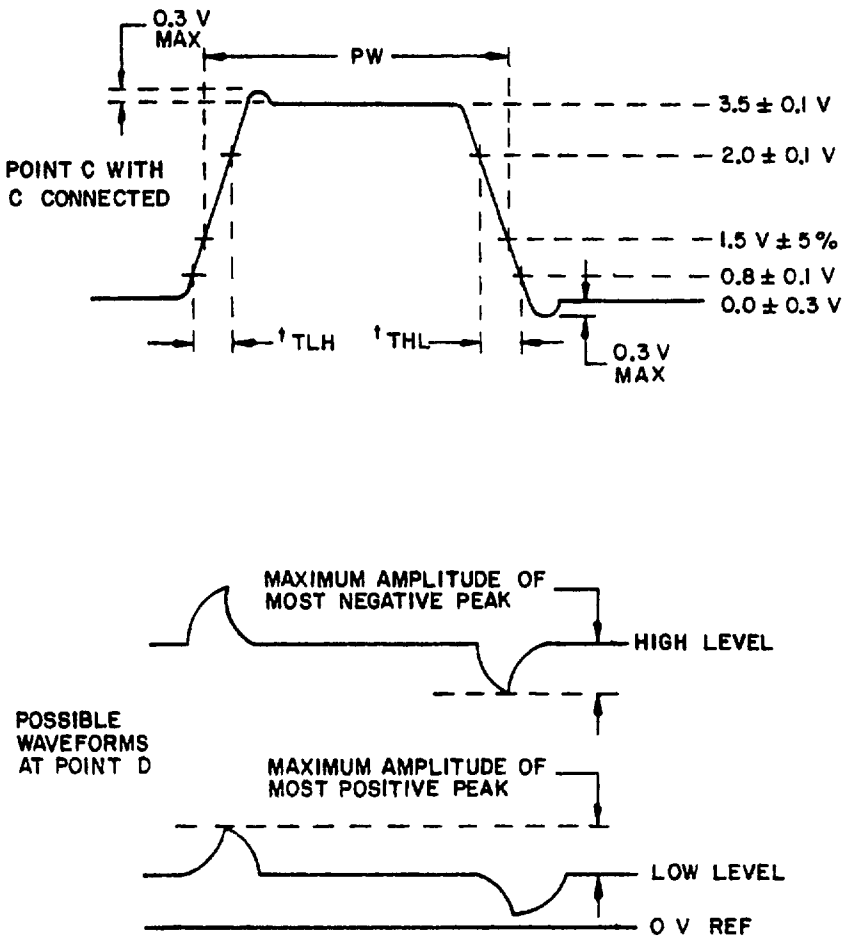
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NOTES
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a load as specified on figure 10.

FIGURE 20. Output crosstalk (high-level and low-level) test circuit for table XV.

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NOTE: Input pulse parameters.
PW = 200 ±20 ns.
PRR = 1.0 MHz ±10%.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

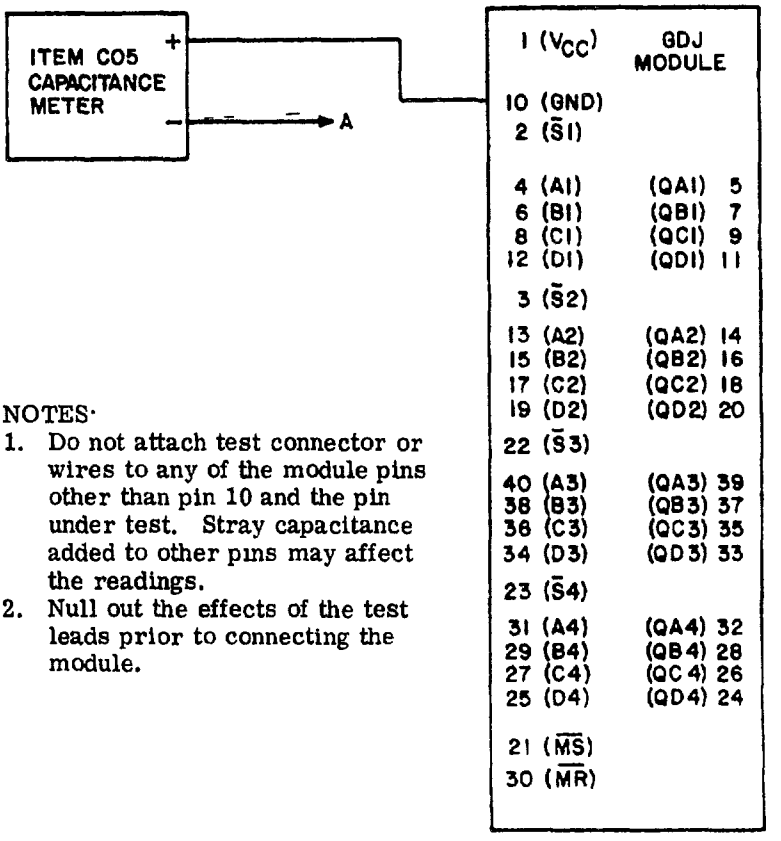
FIGURE 21. Output crosstalk (high-level and low-level) test circuit waveforms for table XV.

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TABLE XVI Input capacitance test

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
X1	A1 Input capacitance	4	2	15	pF	22
X2	B1	6				
X3	C1	8				
X4	D1	12				
X5	A2	13				
X6	B2	15				
X7	C2	17				
X8	D2	19				
X9	A3	40				
X10	B3	38				
X11	C3	36				
X12	D3	34				
X13	A4	31				
X14	B4	29				
X15	C4	27				
X16	D4	25				
X17	$\overline{S}1$	2	3	22		
X18	$\overline{S}2$	3				
X19	$\overline{S}3$	22				
X20	$\overline{S}4$	23				
X21	$\overline{MS}$	21	8	60		
X22	$\overline{MR}$	30	8	60		

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

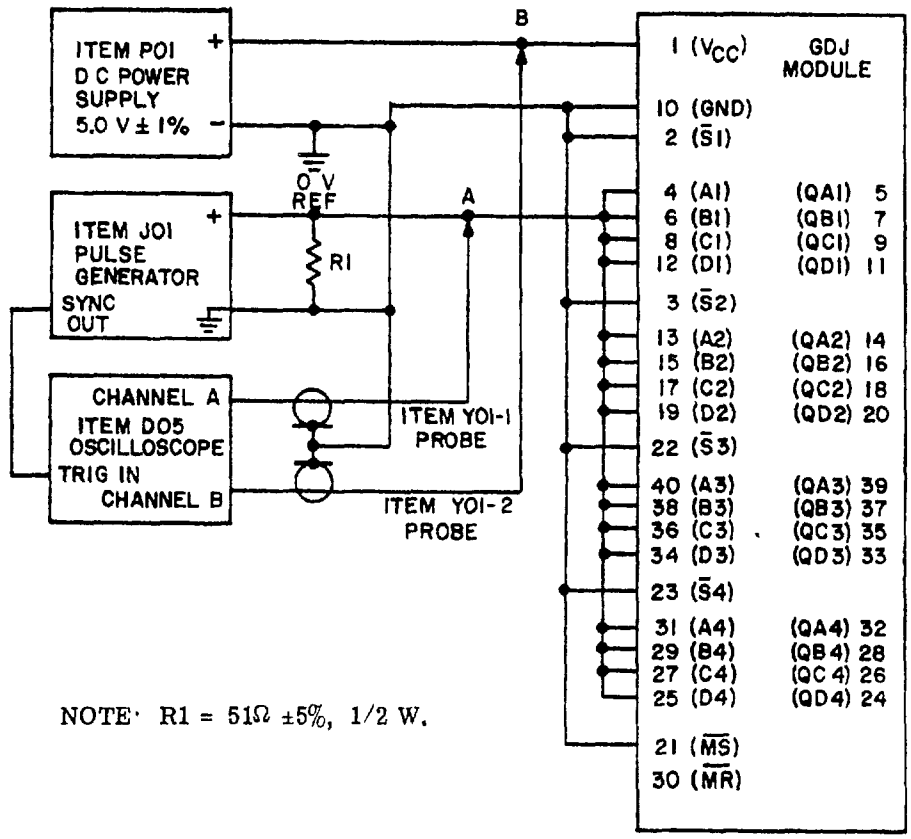
FIGURE 22. Input capacitance test circuit for table XVI.

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TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient amplitude	0	600	mV	23, 24

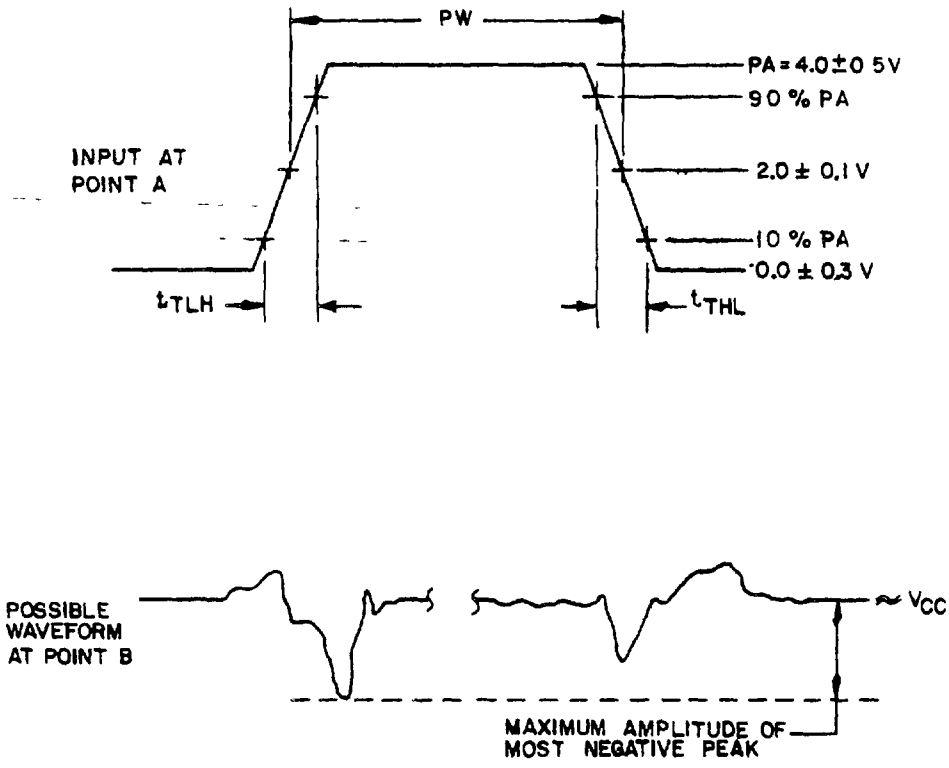
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 23. Power supply transient amplitude test circuit for table XVII.

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NOTE Input pulse parameters
PW = 3.0 ±0.3 μs.
PRR = 100 ±10 kHz.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

* FIGURE 24 Power supply transient amplitude test circuit waveforms for table XVII

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This amendment forms a part of MIL-M-28787/1C(EC), dated 5 May 1982, and is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, AND 6

TABLE I, class II heading: Delete "100° ± 5°C" and substitute "85°C ± 5°C".

PAGE 7

* 4.4.2, delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. Test conditions A, B, C, F, and U are stress tests and will be performed only at 25°C ±5°C. If test conditions A, B, C, F, or U are part of a sequence of tests, the order of testing shall be as follows: U, A, B, C, F where applicable, followed by the remaining tests in the sequence."

* 4.5.1, fifth line: Delete "C" and substitute "D, E, G".

* 4.5.2, fourth line: Add after the word column "(except test condition F)".

* 4.5.3, third line: Delete "C" and substitute "D, E, G".

* 4.5.3, sixth line: Delete "C" and substitute "D, E, G".

PAGE 8

6.4, delete and substitute the following:

"6.4 Cross reference of classification. Military Specification MIL-M-28787/1 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658643. Part number M28787/1-1 supersedes part numbers 2658643 and M28787/1."

6.5, second line: Delete "100°C" and substitute "85°C".

PAGE 9

6.6.4.2, last sentence: Delete and substitute "The preceding relationships also hold for circuits 2-4 (see figure 1)."

PAGE 14

FIGURE 5: Delete and substitute new figure 5 as printed on page 3 of this amendment.

PAGE 18

FIGURE 7: Delete and substitute new figure 7 as printed on page 4 of this amendment.

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FIGURE 9: Delete and substitute new figure 9 as printed on page 5 of this amendment.

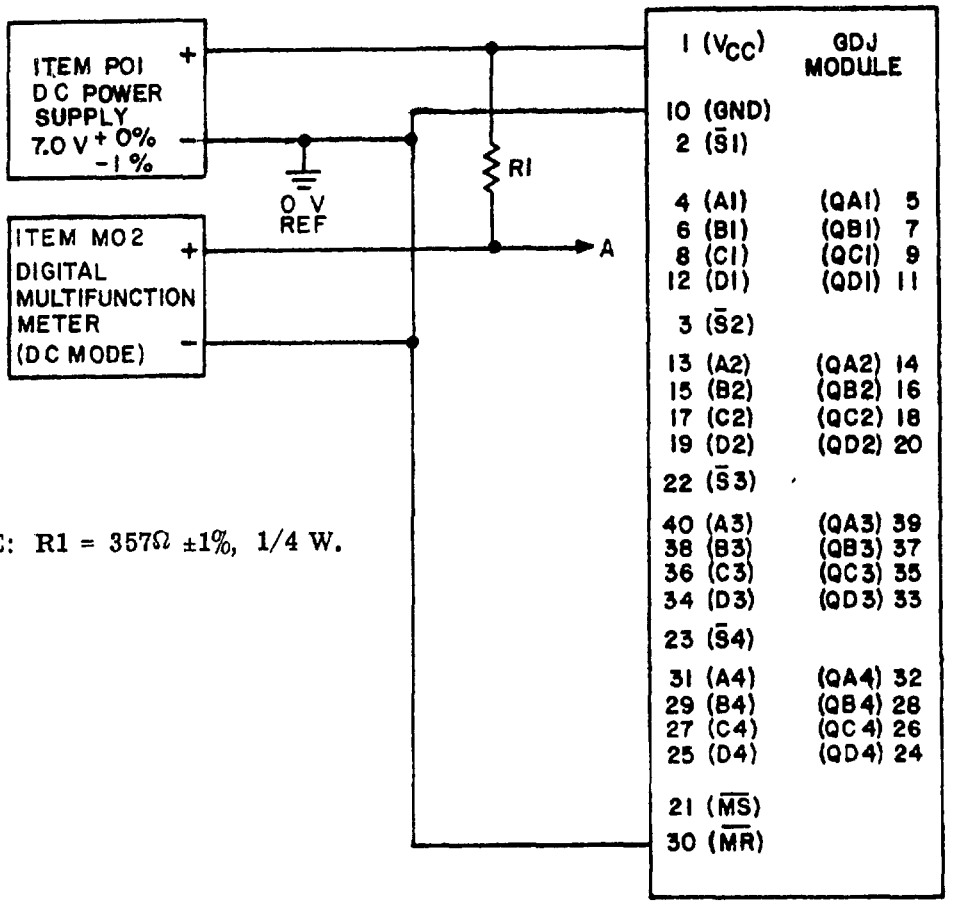
PAGE 30

FIGURE 10: Delete and substitute new figure 10 as printed on page 6 of this amendment.

PAGE 65

* FIGURE 24: Delete and substitute new figure 24 as printed on page 7 of this amendment.

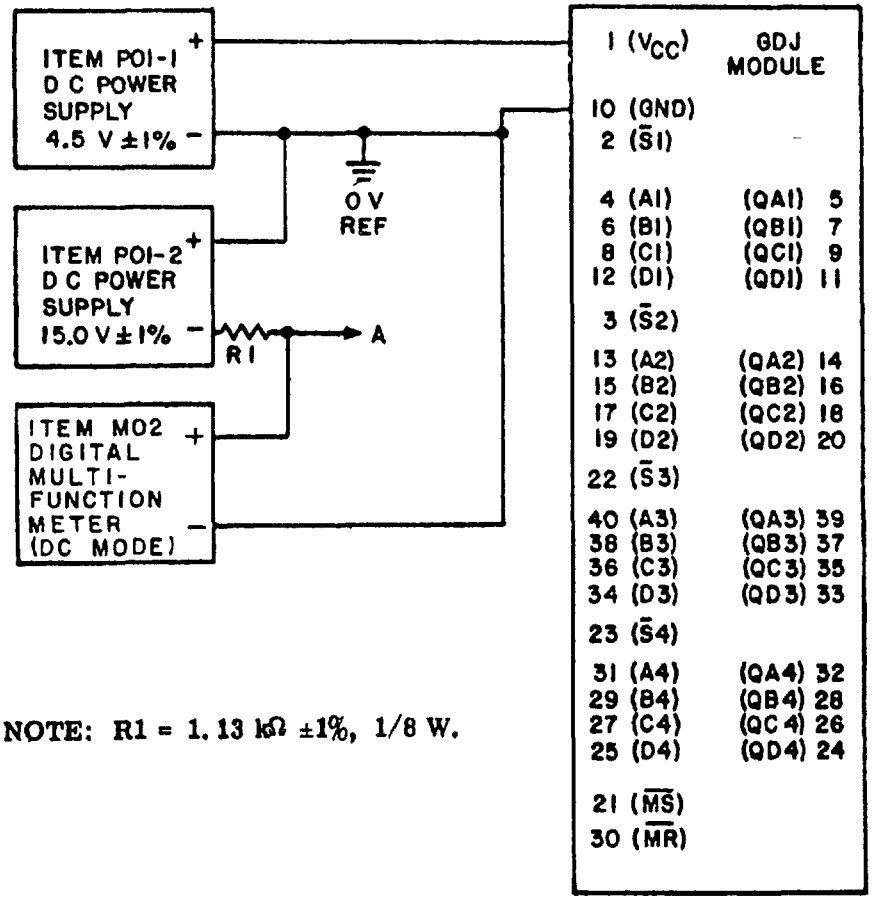
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NOTE: R1 = 357Ω ±1%, 1/4 W.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

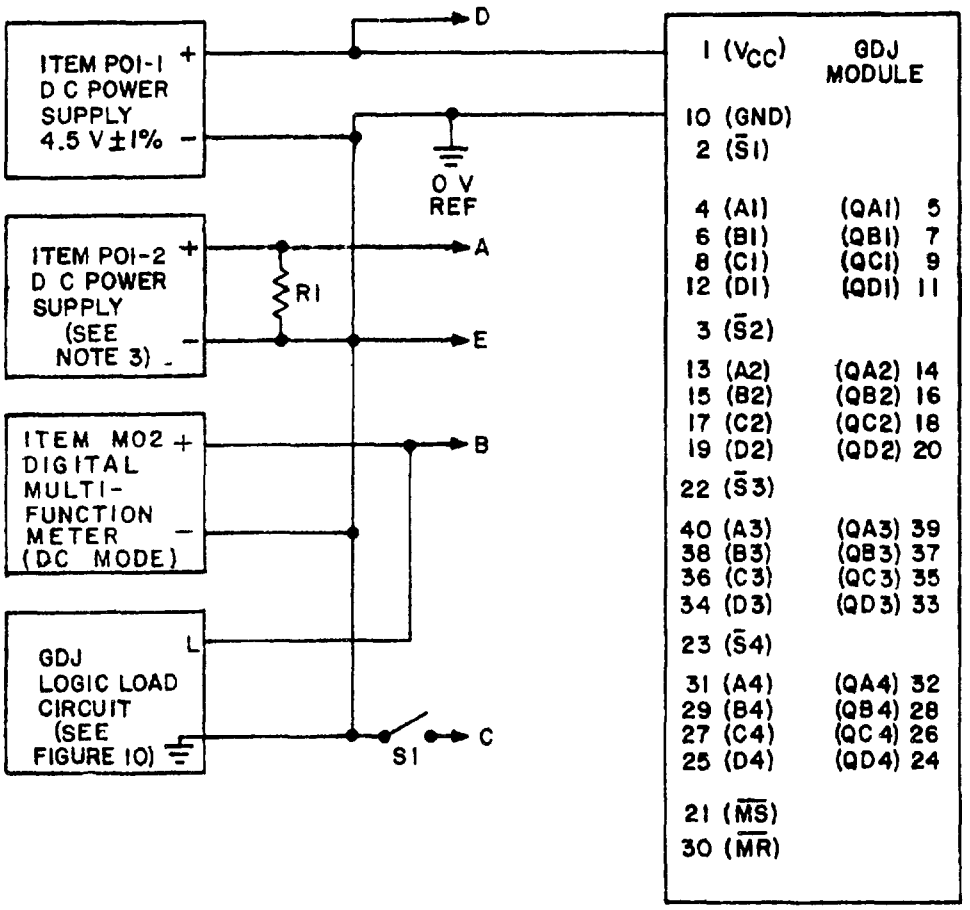
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NOTE: R1 = 1.13 kΩ ±1%, 1/8 W.

FIGURE 7. Input clamp diode voltage test circuit for table V.

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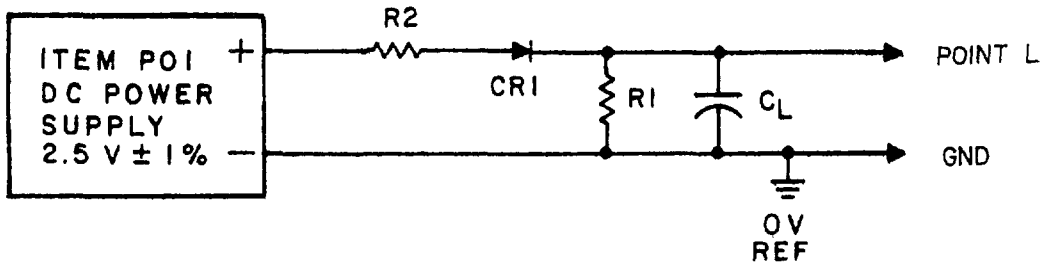
NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. S1 = SPST switch.
3. Adjust item P01-2 as follows:

Test condition	Voltage
G1-G16, G33-G48, H17-H48	2.0 V ±1%
G17-G32, H1-H16, H49-H64	0.8 V ±1%

4. After making all circuit connections, close switch S1 for a minimum of 100 ns; then open switch S1 and allow a minimum of 50 ns before recording measurement.

FIGURE 9. Output voltage (high-level and low-level) test circuit for table VII.

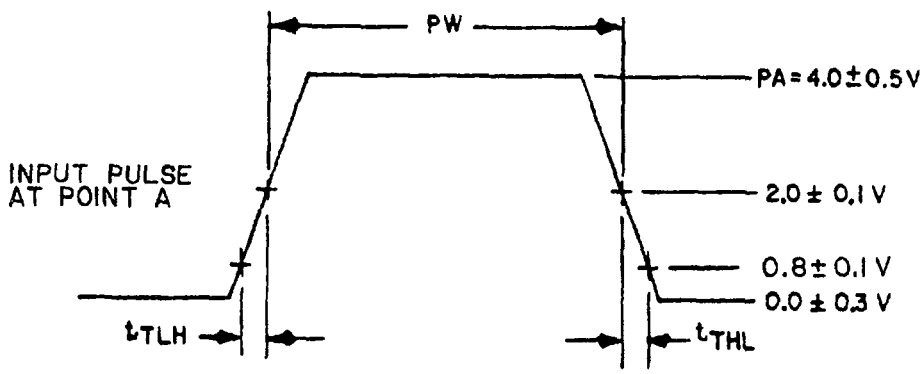
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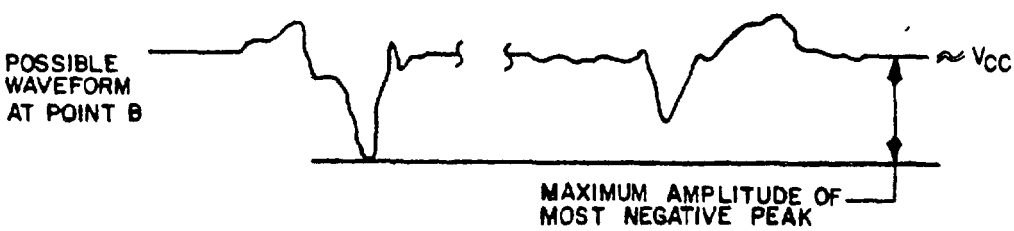


NOTES:
1. CR1 = 1N4150, or equivalent.
2. C_L = 130 ±5 pF (including probe and parasitic capacitance) for test conditions K, L, M, N, P, R, and S.
3. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions V and W.
4. R1 = 4.02 kΩ ±1%, 1/8 W.
5. R2 = Select value for a current flow of 14.4 mA ±1% out of point L with point L held at 0 4 V.
6. Capacitor C_L is not required for dc electrical test conditions G and H.
7. During output switching, item P01 shall have a maximum overshoot of 500 mV pk.

FIGURE 10. GDJ logic load circuit.

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NOTE: Input pulse parameters:
PW = 3.0 ±0.3 μs.
PRR = 100 ±10 kHz.
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVII.

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